

BOJNORD, Iran, Islamic Republic ofWMO: **407230**Lat: **37.493N**Lon: **57.308E**Elev: **1080**StdP: **89.01**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.1	-15.2	1.1	-7.5	-12.3	1.5	-3.7	11.1	5.9	9.7	8.4	0.5	270	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	36.0	17.9	34.2	17.4	33.0	17.2	21.6	29.8	20.7	28.8	19.8	27.8	3.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	15.9	25.4	18.1	14.8	24.7	17.1	13.9	24.1	68.5	29.6	64.9	28.8	61.8	27.7	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.3	7.2	-13.2	38.7	3.5	1.5	-15.7	39.8	-17.8	40.7	-19.8	41.5	-22.3	42.6		
(5)			-13.8	23.1	4.0	1.1	-16.6	23.9	-19.0	24.6	-21.2	25.2	-24.1	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	1.3	3.2	8.4	13.1	18.9	23.4	25.9	24.5	20.7	14.5	7.3	3.3
	DBStd	9.30	4.62	4.76	4.41	4.31	2.81	2.86	2.34	2.31	2.78	3.68	4.02	3.72
	HDD10.0	883	271	194	83	19	0	0	0	0	0	7	98	211
	HDD18.3	2389	528	424	309	162	27	3	0	0	11	127	331	467
	CDD10.0	2257	1	3	33	114	277	402	492	449	322	146	17	2
	CDD18.3	722	0	0	1	6	46	154	234	191	82	8	0	0
	CDH23.3	8496	0	1	18	109	578	1786	2686	2266	901	148	3	0
	CDH26.7	3892	0	0	3	22	178	849	1387	1101	322	31	0	0
Wind	WSAvg	2.2	1.5	1.8	2.4	2.3	2.5	3.0	3.2	2.7	2.3	1.8	1.2	1.3
Precipitation	PrecAvg	303	37	37	58	46	30	12	4	3	5	11	29	30
	PrecMax	444	64	71	115	93	66	35	29	21	31	36	65	57
	PrecMin	179	15	14	19	4	1	0	0	0	0	0	1	7
	PrecStd	56	14	16	25	23	16	9	7	5	8	10	17	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.1	25.4	29.2	32.9	36.9	38.9	37.7	34.3	30.0	23.2	17.2
		MCWB	6.6	9.8	11.8	15.1	16.4	17.3	18.7	17.9	16.4	14.8	11.8	7.3
	2%	DB	13.8	15.8	22.0	26.2	30.4	34.9	36.8	35.0	32.1	26.9	19.3	14.1
		MCWB	6.0	7.7	10.7	14.1	16.0	17.1	18.3	17.4	16.1	14.1	10.8	6.7
	5%	DB	11.0	13.0	19.2	23.9	28.7	33.1	34.9	33.5	30.1	24.8	16.9	12.0
		MCWB	5.2	6.5	9.8	13.1	15.6	16.5	18.3	16.9	16.2	13.6	9.8	5.9
	10%	DB	8.3	10.6	16.6	21.2	26.8	31.5	33.1	32.0	28.2	22.5	14.8	9.8
		MCWB	3.8	5.6	9.1	12.3	15.3	16.4	18.2	16.8	15.8	13.0	8.9	5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	10.3	13.1	16.7	18.9	20.9	24.3	22.3	20.7	16.9	12.9	9.1
		MCDB	13.5	17.3	21.8	24.6	27.4	30.5	32.6	30.4	28.2	25.2	19.9	15.2
	2%	WB	6.6	8.5	11.7	15.2	17.7	19.6	22.0	21.2	19.0	15.7	11.7	7.7
		MCDB	12.4	14.3	19.6	23.8	26.6	28.6	30.2	28.8	26.3	23.2	18.1	12.1
	5%	WB	5.4	7.2	10.6	14.1	16.9	18.9	21.1	20.1	18.2	14.8	10.5	6.6
		MCDB	10.3	12.2	17.8	21.7	25.3	28.0	29.6	27.8	25.3	21.9	15.5	10.6
	10%	WB	4.1	5.9	9.4	13.0	16.1	18.2	20.4	19.1	17.3	13.9	9.3	5.4
		MCDB	7.8	10.0	15.6	19.7	24.0	27.1	28.5	27.0	24.5	20.3	13.8	9.0
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.8	11.9	12.4	13.1	14.0	13.7	14.9	14.6	14.0	11.9	10.3
		MCDBR	13.7	14.8	16.5	16.8	17.2	17.3	16.9	17.3	17.8	17.8	16.1	13.8
		MCWBR	8.0	8.4	7.6	7.4	6.6	6.0	6.0	6.9	7.8	8.4	9.2	8.2
	5% WB	MCDBR	12.4	13.3	15.1	14.8	14.6	13.5	13.3	13.4	13.7	14.9	13.6	11.8
		MCWBR	7.6	7.9	7.8	7.0	6.5	5.3	5.2	5.8	6.4	7.3	8.1	7.7
Clear-Sky Solar Irradiance	taub		0.330	0.372	0.450	0.503	0.515	0.497	0.507	0.454	0.418	0.424	0.367	0.327
	taud		2.257	2.123	1.919	1.842	1.849	1.905	1.883	2.011	2.086	2.035	2.174	2.278
	Ebn at Noon		838	841	802	785	783	796	783	818	826	770	780	813
	Edh at Noon		101	131	177	203	205	194	197	169	148	141	109	92
All-Sky Solar Radiation	RadAvg		2.27	3.03	4.20	5.47	6.66	7.84	7.70	7.20	5.90	4.18	2.79	2.06
	RadStd		0.20	0.30	0.33	0.54	0.33	0.22	0.21	0.33	0.20	0.21	0.19	0.24